

Shahriar Alom Masud

Embedded Systems & Edge AI Engineer

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Professional Summary

IoT and Embedded Systems Engineer and author of SmartProv (v2.1.3), a published open-source ESP32 library on the Arduino Library Manager. Experienced in C/C++ firmware development, FreeRTOS multitasking, wireless telemetry (LoRa, MQTT), and deploying lightweight Edge AI models to microcontrollers. Delivers end-to-end connected systems from sensor-level interfacing and fail-safe OTA update pipelines to real-time cloud analytics and operational web dashboards.

Technical Skills

Programming Languages: C, C++, Python, TypeScript, JavaScript, SQL, Bash

Embedded Systems & Firmware: ESP32, ESP8266, ESP32-CAM, Raspberry Pi, Arduino, FreeRTOS, Dual-Partition OTA Rollback, Hardware Timers, ISR, Memory Management (NVS/EEPROM), Wi-Fi Provisioning

Protocols & Hardware Interfacing: LoRa, MQTT, UART, SPI, I2C, 1-Wire, WebSockets, Logic-Level Shifting, Multi-Sensor Integration, Relay & Motor Drivers

Edge AI, TinyML & Computer Vision: TinyML, Edge AI, ONNX Runtime, OpenCV, YOLOv8, MediaPipe, Explainable AI (TreeSHAP, Grad-CAM), PyTorch, Scikit-learn, Time-Series Forecasting, LLM Integration

Cloud, Backend & Web: AWS, FastAPI, InfluxDB Cloud, Supabase (PostgreSQL), Firebase Realtime DB, SQLite, React 18/19, Tailwind CSS, Vite, Chart.js

DevOps, Tools & Automation: Git, GitHub Actions (CI/CD), Docker, Linux / Shell, Postman, n8n, PlatformIO, Arduino IDE, PyInstaller, Serial Debugging

Experience

Open Source Developer — SmartProv

Apr 2026 – Present

github.com/Masud744/SmartProv — Arduino Library Manager, ESP32, ESP8266, C++, NVS/EEPROM

- Authored and published **SmartProv (v2.1.3)** on the official Arduino Library Manager, providing an asynchronous captive-portal Wi-Fi provisioning framework for ESP32 and ESP8266 microcontrollers.
- Implemented non-volatile flash persistence via NVS/EEPROM with multi-network fallback and factory-reset routines, eliminating manual Wi-Fi credential flashing across 100% of device deployments.
- Adopted by embedded developers globally; reduced initial device network configuration time from minutes to under 30 seconds.

Technical Projects

RespiGuard — Explainable Edge AI Respiratory Risk Monitoring

May 2026 – Sep 2026

github.com/Masud744/RespiGuard — Live: respiuard-fvh9.onrender.com

ESP32, PMS5003, MQ-135, DHT22, FreeRTOS, FastAPI, TreeSHAP, React, Vite, Supabase, Groq LLaMA 3.3, Render

- Designed an ESP32 edge sensing node (PMS5003 laser particulate counter, MQ-135, DHT22) running FreeRTOS firmware to collect localized air quality telemetry across 5 atmospheric parameters.
- Deployed a hierarchical ML model with dynamic TreeSHAP explainability for asthma triage, achieving sub-100ms local inference; built real-time React 18 dashboard on Render with geospatial risk maps and Groq LLaMA-3.3 copilot.

Solar-Aware HEMS — Risk-Aware Smart Home Energy Management

Jan 2026 – Sep 2026

github.com/Masud744/solar-aware-hems — Live: solar-aware-hems.vercel.app

Python, FastAPI, Scikit-learn, TreeSHAP, Open-Meteo, Supabase, ESP32, C++, React, Plotly, Vercel

- Engineered risk-aware HEMS combining ML solar and load forecasting with empirical uncertainty ($k\sigma$ safety margins) to calculate Safe Solar Surplus across 24-hour cycles.
- Constructed ESP32 hardware layer for AC voltage/current telemetry and 8-channel relay scheduling, achieving automated peak-load shaving and zero grid overdraw during peak tariff intervals.

ESP32 SmartProv GitHub OTA Update System

Jun 2026 – Aug 2026

github.com/Masud744/esp32-smartprov-github-ota

ESP32, C/C++, FreeRTOS, GitHub REST API, HTTPS, NVS, Dual-Partition OTA

- Architected dual-partition flash update mechanism (ota_0/ota_1) polling GitHub Releases API over secure HTTPS with semantic version verification.
- Engineered fail-safe rollback logic using NVS boot counters and watchdog timers; verified across **15+ physical ESP32 nodes** with 0% bricking rate under simulated power and network drops.

Education

University of Frontier Technology, Bangladesh

Aug 2023 – Jun 2027 (Expected)

Bachelor of Science (Engineering) in Internet of Things & Robotics Engineering

Gazipur, Bangladesh

Leadership & Activities

Head of Membership Development — University Robotics Club

Oct 2025 – Present

Organized technical workshops on ESP32 firmware and IoT interfacing for 60+ junior engineering students; mentored members in microcontroller programming and circuit design.

RoboFusion 1.0 — Robo Soccer Segment Leader

Aug 2026

Coordinated arena technical setup, wireless frequency allocation, and referee operations for inter-university robot soccer matches.